# RENEWABLE ENERGY AND GREEN HYDROGEN

**POWER METRIC FIREWALL → ENERGY CATEGORY MAP → GRID INTEGRATION RAIL → PROJECT STATUS LADDER → ELECTROLYSIS PATHWAY → HYDROGEN CLAIM STACK**

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure: the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◉ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

## STAGE 00 POWER METRIC FIREWALL

POWER METRIC FIREWALL · INSTALLED CAPACITY · NAMEPLATE POWER CAPABILITY · ELECTRICAL ENERGY PRODUCED OVER TIME · CAPACITY SHARE · PORTFOLIO CAPABILITY, NOT OUTPUT SHARE · RELIABLE SUPPLY · TIMING, FLEXIBILITY AND NETWORK MATTER

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| INSTALLED CAPACITY → nameplate power capability | CAPACITY SHARE → portfolio capability, not output share | RULE → never convert megawatts into generation without evidence |
| GENERATION → electrical energy produced over time | RELIABLE SUPPLY → timing, flexibility and network matter | MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,... |

**CORE CLAIM**

- INSTALLED CAPACITY → nameplate power capability
- GENERATION → electrical energy produced over time

**MECHANISM**

- CAPACITY SHARE → portfolio capability, not output share
- RELIABLE SUPPLY → timing, flexibility and network matter

**CONSEQUENCE / LIMIT**

- RULE → never convert megawatts into generation without evidence
- MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert megawatts into generation without evidence.**

01

## STAGE 01 ENERGY CATEGORY MAP

ENERGY CATEGORY MAP · SOLAR WIND HYDRO BIOMASS GEOTHERMAL · RENEWABLE SOURCES PLUS OTHER NON-FOSSIL SOURCES · GENERATED ENERGY OUTPUT · MANUFACTURED ENERGY CARRIER · SOURCE, OUTPUT AND CARRIER ARE DIFFERENT CATEGORIES

**RENEWABLE → solar wind hydro biomass geothermal**

NON-FOSSIL → renewable sources plus other non-fossil sources → ELECTRICITY → generated energy output → HYDROGEN → manufactured energy carrier

**CORE CLAIM**

- RENEWABLE → solar wind hydro biomass geothermal
- NON-FOSSIL → renewable sources plus other non-fossil sources

**MECHANISM**

- ELECTRICITY → generated energy output
- HYDROGEN → manufactured energy carrier

**CONSEQUENCE / LIMIT**

- RULE → source, output and carrier are different categories
- RENEWABLE → solar wind hydro biomass geothermal

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → source, output and carrier are different categories.**

02

## STAGE 02 GRID INTEGRATION RAIL

GRID INTEGRATION RAIL · VARIABLE GENERATION · FORECAST RESOURCE-DEPENDENT OUTPUT · MOVE POWER FROM RESOURCE TO DEMAND · FLEXIBLE DEMAND AND BALANCING · MATCH SYSTEM CONDITIONS · SHIFT ENERGY AND PROVIDE SELECTED SERVICES · RELIABILITY IS A SYSTEM OUTCOME

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| VARIABLE GENERATION → forecast resource-dependent output | FLEXIBLE DEMAND AND BALANCING → match system conditions | DELIVERY → reliability is a system outcome |
| TRANSMISSION → move power from resource to demand | STORAGE → shift energy and provide selected services | VARIABLE GENERATION → forecast resource-dependent output |

**CORE CLAIM**

- VARIABLE GENERATION → forecast resource-dependent output
- TRANSMISSION → move power from resource to demand

**MECHANISM**

- FLEXIBLE DEMAND AND BALANCING → match system conditions
- STORAGE → shift energy and provide selected services

**CONSEQUENCE / LIMIT**

- DELIVERY → reliability is a system outcome
- VARIABLE GENERATION → forecast resource-dependent output

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE → shift energy and provide selected services.**

03

## STAGE 03 PROJECT STATUS LADDER

PROJECT STATUS LADDER · TARGET OR ANNOUNCEMENT · INTENDED FUTURE SCALE · ALLOCATION OR AWARD · CONTRACTUAL OR POLICY COMMITMENT · PHYSICAL IMPLEMENTATION STAGE · ASSET ENTERS SERVICE · GENERATION OR PRODUCTION

**TARGET OR ANNOUNCEMENT → intended future scale**

ALLOCATION OR AWARD → contractual or policy commitment → CONSTRUCTION → physical implementation stage → COMMISSIONING → asset enters service

**CORE CLAIM**

- TARGET OR ANNOUNCEMENT → intended future scale
- ALLOCATION OR AWARD → contractual or policy commitment

**MECHANISM**

- CONSTRUCTION → physical implementation stage
- COMMISSIONING → asset enters service

**CONSEQUENCE / LIMIT**

- GENERATION OR PRODUCTION → measured operating output
- TARGET OR ANNOUNCEMENT → intended future scale

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALLOCATION OR AWARD → contractual or policy commitment.**

04

## STAGE 04 ELECTROLYSIS PATHWAY

ELECTROLYSIS PATHWAY · ELECTRICITY SOURCE · ATTRIBUTE MUST BE EVIDENCED · WATER AND TREATMENT · PROJECT-SPECIFIC INPUT CONTEXT · WATER SPLIT INTO HYDROGEN AND OXYGEN · HYDROGEN OUTPUT · CHEMICAL PRODUCT

ELECTRICITY SOURCE → attribute must be evidenced → WATER AND TREATMENT → project-specific input context → ELECTROLYSER → water split into hydrogen and oxygen → HYDROGEN OUTPUT → chemical product → GREEN CLAIM → standard and certification layer still required

**CORE CLAIM**

- ELECTRICITY SOURCE → attribute must be evidenced
- WATER AND TREATMENT → project-specific input context

**MECHANISM**

- ELECTROLYSER → water split into hydrogen and oxygen
- HYDROGEN OUTPUT → chemical product

**CONSEQUENCE / LIMIT**

- GREEN CLAIM → standard and certification layer still required
- ELECTRICITY SOURCE → attribute must be evidenced

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GREEN CLAIM → standard and certification layer still required.**

05

## STAGE 05 HYDROGEN CLAIM STACK

HYDROGEN CLAIM STACK · PRODUCTION PATHWAY · HOW HYDROGEN IS MADE · LIFECYCLE BOUNDARY · INCLUDED EMISSIONS STAGES · OFFICIAL STANDARD · QUALIFYING CONDITIONS · DOCUMENTED CONFORMITY

PRODUCTION PATHWAY → how hydrogen is made → LIFECYCLE BOUNDARY → included emissions stages → OFFICIAL STANDARD → qualifying conditions → CERTIFICATION → documented conformity → MARKET LABEL → cannot replace the layers below

**CORE CLAIM**

- PRODUCTION PATHWAY → how hydrogen is made
- LIFECYCLE BOUNDARY → included emissions stages

**MECHANISM**

- OFFICIAL STANDARD → qualifying conditions
- CERTIFICATION → documented conformity

**CONSEQUENCE / LIMIT**

- MARKET LABEL → cannot replace the layers below
- PRODUCTION PATHWAY → how hydrogen is made

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MARKET LABEL → cannot replace the layers below.**

06

## STAGE 06 USE-CASE DECISION TREE

USE-CASE DECISION TREE · CAN CLEAN ELECTRICITY BE USED DIRECTLY · PREFER DIRECT ROUTE · MOLECULE OR HIGH HEAT NEEDED · TEST HYDROGEN FIT · SECTOR AND DUTY CYCLE · IDENTIFY TECHNICAL CONSTRAINT · CONVERSION LOSS AND INFRASTRUCTURE

**CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route**

MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit → SECTOR AND DUTY CYCLE → identify technical constraint → CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

**CORE CLAIM**

- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route
- MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

**MECHANISM**

- SECTOR AND DUTY CYCLE → identify technical constraint
- CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

**CONSEQUENCE / LIMIT**

- VERDICT → hydrogen is selective, not universal
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR AND DUTY CYCLE → identify technical constraint.**

07

## STAGE 07 DERIVATIVE CONVERSION MAP

DERIVATIVE CONVERSION MAP · BASE ENERGY CARRIER AND FEEDSTOCK · NITROGEN-BASED DERIVATIVE · METHANOL OR SYNTHETIC FUEL · CARBON-CONTAINING DERIVATIVE · CONVERSION AND DELIVERY · EXTRA ENERGY AND INFRASTRUCTURE · PRODUCT TARGETS AND OUTPUTS REMAIN SEPARATE

**HYDROGEN → base energy carrier and feedstock**

AMMONIA → nitrogen-based derivative → METHANOL OR SYNTHETIC FUEL → carbon-containing derivative → CONVERSION AND DELIVERY → extra energy and infrastructure

**CORE CLAIM**

- HYDROGEN → base energy carrier and feedstock
- AMMONIA → nitrogen-based derivative

**MECHANISM**

- METHANOL OR SYNTHETIC FUEL → carbon-containing derivative
- CONVERSION AND DELIVERY → extra energy and infrastructure

**CONSEQUENCE / LIMIT**

- RULE → product targets and outputs remain separate
- HYDROGEN → base energy carrier and feedstock

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → product targets and outputs remain separate.**

08

## STAGE 08 WATER AND SITE SCREEN

WATER AND SITE SCREEN · FRESHWATER TREATED WATER OR OTHER VERIFIED SUPPLY · PRETREATMENT REQUIREMENT · BASIN AND LOCAL SCARCITY CONTEXT · COMPETING DEMAND · ECOLOGICAL AND SOCIAL CLAIMS · PROJECT EVIDENCE, NOT A UNIVERSAL WATER NUMBER

**SOURCE → freshwater treated water or other verified supply**

QUALITY → pretreatment requirement → LOCATION → basin and local scarcity context → COMPETING DEMAND → ecological and social claims

**CORE CLAIM**

- SOURCE → freshwater treated water or other verified supply
- QUALITY → pretreatment requirement

**MECHANISM**

- LOCATION → basin and local scarcity context
- COMPETING DEMAND → ecological and social claims

**CONSEQUENCE / LIMIT**

- VERDICT → project evidence, not a universal water number
- SOURCE → freshwater treated water or other verified supply

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → project evidence, not a universal water number.**

09

## STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN · HYDROGEN LEAVES ELECTROLYSER · COMPRESSION LIQUEFACTION OR CONVERSION · STORAGE AND TRANSPORT · VESSEL PIPELINE TANKER OR PORT · END USE · EQUIPMENT AND SAFETY COMPATIBILITY · DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser → CONDITIONING → compression liquefaction or conversion → STORAGE AND TRANSPORT → vessel pipeline tanker or port → END USE → equipment and safety compatibility → OUTCOME → delivered service differs from production capacity

**CORE CLAIM**

- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

**MECHANISM**

- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

**CONSEQUENCE / LIMIT**

- OUTCOME → delivered service differs from production capacity
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free.**

10

## STAGE 10 MISSION AND SIGHT MAP

MISSION AND SIGHT MAP · MISSION TARGET · INTENDED FUTURE PRODUCTION SCALE · SIGHT MANUFACTURING · ELECTROLYSER CAPABILITY SUPPORT · SIGHT PRODUCTION · HYDROGEN OUTPUT SUPPORT · COMMISSIONED OUTPUT

| CORE CLAIM | MECHANISM | CONSEQUENCE / LIMIT |
|---|---|---|
| MISSION TARGET → intended future production scale | SIGHT PRODUCTION → hydrogen output support | CLIMATE OUTCOME → separate lifecycle result |
| SIGHT MANUFACTURING → electrolyser capability support | COMMISSIONED OUTPUT → later measured evidence | EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,... |

**CORE CLAIM**

- MISSION TARGET → intended future production scale
- SIGHT MANUFACTURING → electrolyser capability support

**MECHANISM**

- SIGHT PRODUCTION → hydrogen output support
- COMMISSIONED OUTPUT → later measured evidence

**CONSEQUENCE / LIMIT**

- CLIMATE OUTCOME → separate lifecycle result
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

**MECHANISM / VERDICT —** EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,.**

11

## STAGE 11 ENERGY-TRANSITION ANSWER SPINE

ENERGY-TRANSITION ANSWER SPINE · CAPACITY GENERATION RELIABILITY SOURCE AND CARRIER · GRID INTEGRATION ELECTROLYSIS CERTIFICATION AND DELIVERY · DIRECT ELECTRIFICATION BEFORE SELECTIVE HYDROGEN USE · TARGET AWARD COMMISSION OUTPUT AND OUTCOME · DATED MNRE POWER-SYSTEM AND STANDARDS EVIDENCE

DEFINE → capacity generation reliability source and carrier → TRACE → grid integration electrolysis certification and delivery → SCREEN → direct electrification before selective hydrogen use → SEPARATE → target award commission output and outcome → VERIFY → dated MNRE power-system and standards evidence

**CORE CLAIM**

- DEFINE → capacity generation reliability source and carrier
- TRACE → grid integration electrolysis certification and delivery

**MECHANISM**

- SCREEN → direct electrification before selective hydrogen use
- SEPARATE → target award commission output and outcome

**CONSEQUENCE / LIMIT**

- VERIFY → dated MNRE power-system and standards evidence
- DEFINE → capacity generation reliability source and carrier

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated MNRE power-system and standards evidence.**

E

## EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · LARGE-SCALE SOLAR AND WIND INFRASTRUCTURE REQUIRES SUBSTANTIAL LAND AREA · BALANCED FRAMING · THIS IS NOT A CASE WHERE EITHER RENEWABLE ENERGY · BATTERY-STORAGE COSTS HAVE BEEN FALLING BUT REMAIN A SUBSTANTIAL · ANALYTICAL POINT · INDIA'S RENEWABLE-CAPACITY-ADDITION TARGETS (500 GW NON-FOSSIL BY · GREEN-HYDROGEN PRODUCTION-COST FIGURES AND ELECTROLYSER-CAPACITY FIGURES CHANGE RAPIDLY

**OPTIONAL SOURCE DEPTH**

- CAUTION: Large-scale solar and wind infrastructure requires substantial land area and, for wind
- CAUTION: Balanced framing: this is not a case where either renewable energy or species
- CAUTION: Battery-storage costs have been falling but remain a substantial cost component of a
- CAUTION: Analytical point: India's renewable-capacity-addition targets (500 GW non-fossil by

**ADVANCED DEBATE**

- CAUTION: Green-hydrogen production-cost figures and electrolyser-capacity figures change rapidly
- CAUTION: Progress toward the National Green Hydrogen Mission's 5 MMT/year 2030 target and the
- CAUTION: Quantifying the precise trade- off between renewable-infrastructure expansion and
- Solar and wind generation are variable/non-dispatchable, requiring storage and grid

**LEGACY / NUANCE**

- Green hydrogen's cost-competitiveness depends on both falling electrolyser/renewable
- The Supreme Court has engaged with the tension between renewable-energy transmission
- The SIGHT Programme's dual incentive structure (electrolyser manufacturing plus
- NOT: India's renewable-energy challenge today is primarily about building enough generation

**OPTIONAL STATUS —** This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels